# DisasterM3 vs BRIGHT — Optical Data Audit & EDA

Oct 19, 2026 · @Ridita Afrin

Both datasets are usable for the optical pipeline, but the audit found a validation leak inside DisasterM3, and our five BRIGHT events can add building footprints, not destroyed examples.

- **Validation leak.** 40 of the 645 frozen val pairs (6.2%) are near-copies of train pairs, because xBD's `socal_fire` and `woolsey_fire` share 264 pre images. Those 40 pairs hold 8.6% of the val's destroyed pixels.
- **Our BRIGHT subset does not leak.** None of the 1,285 tiles of our five events is in the optical pool; 1,165 exist in DisasterM3 only as SAR pairs or Bench masks.
- **BRIGHT has no optical post image.** All 4,246 BRIGHT post tiles are SAR, so the 21% destroyed labels of our five events go unused. The tiles enter as intact footprints, 82% of them volcano, while the val holds 8 volcano pairs.
- **The papers' resolutions are wrong.** BRIGHT's GeoTIFF tags contradict its own paper for 6 of 14 events, and DisasterM3's 2,828 BRIGHT copies are 0.30 m, not the stated 0.8 m.
- **Appearance gap.** Our five events are darker (pixel median 60 vs 84) and 2.6× sharper than the pool's post images.
- **Pairs are not "strictly aligned".** 52% of measurable optical pairs are off by more than 2 px, up to 56 px.
- **SAR hides behind file names.** 73% of DisasterM3's SAR posts have no `sar` in the name; the tags and the pixels agree, the names do not.

## Scope and SAR exclusion

Every number below was measured on the local copies of both datasets, and SAR is excluded by pixels, not only by tags.

| Source | Used in this audit | Size |
|---|---|---|
| DisasterM3 Instruct images | pre images + optical post images (pixels) | 20,988 files |
| DisasterM3 Instruct masks | building and road masks of the optical pool | 6,443 images |

| Source | Used in this audit | Size |
|---|---|---|
| DisasterM3 Bench | counts from its JSON only: its images are not in the local copy | 30,042 entries |
| BRIGHT pre-event | optical tiles (pixels, GeoTIFF tags) | 3,395 tiles, 11 events |
| BRIGHT post-event | excluded: all 4,246 tiles are single-band SAR | 4,246 tiles |
| BRIGHT targets | damage label maps (0 background, 1 intact, 2 damaged, 3 destroyed) | 4,246 maps |

- **SAR rule.** An image counts as SAR when its three channels are identical on the pixels that carry data (mean difference under 1 grey level). This is checked against DisasterM3's `image_type` / `post_image_type` tags, because 2,066 DisasterM3 files carry no tag at all.
- **Pairs.** DisasterM3 pre/post pairs come from `train_release.json`: 7,160 optical and 2,648 SAR pairs over 9,117 pre images.
- **Optical pool.** The 6,443 optical Building Damage Assessment pairs the champion notebooks train and validate on, split 5,798 / 645 with `random_state=42` (the frozen E-protocol validation set).
- **BRIGHT's optical side is pre-event only.** The release folder `pre-event_wo_ukraine_myanmar_mexico` leaves out the Google Earth tiles of Ukraine (513), Myanmar (126) and Mexico (212), so those three events have labels and SAR but no optical image here.

## Inventory

By pixel test, the local DisasterM3 Instruct copy holds 17,451 optical and 3,537 SAR images; BRIGHT adds 3,395 optical pre-event tiles and no optical post-event image.

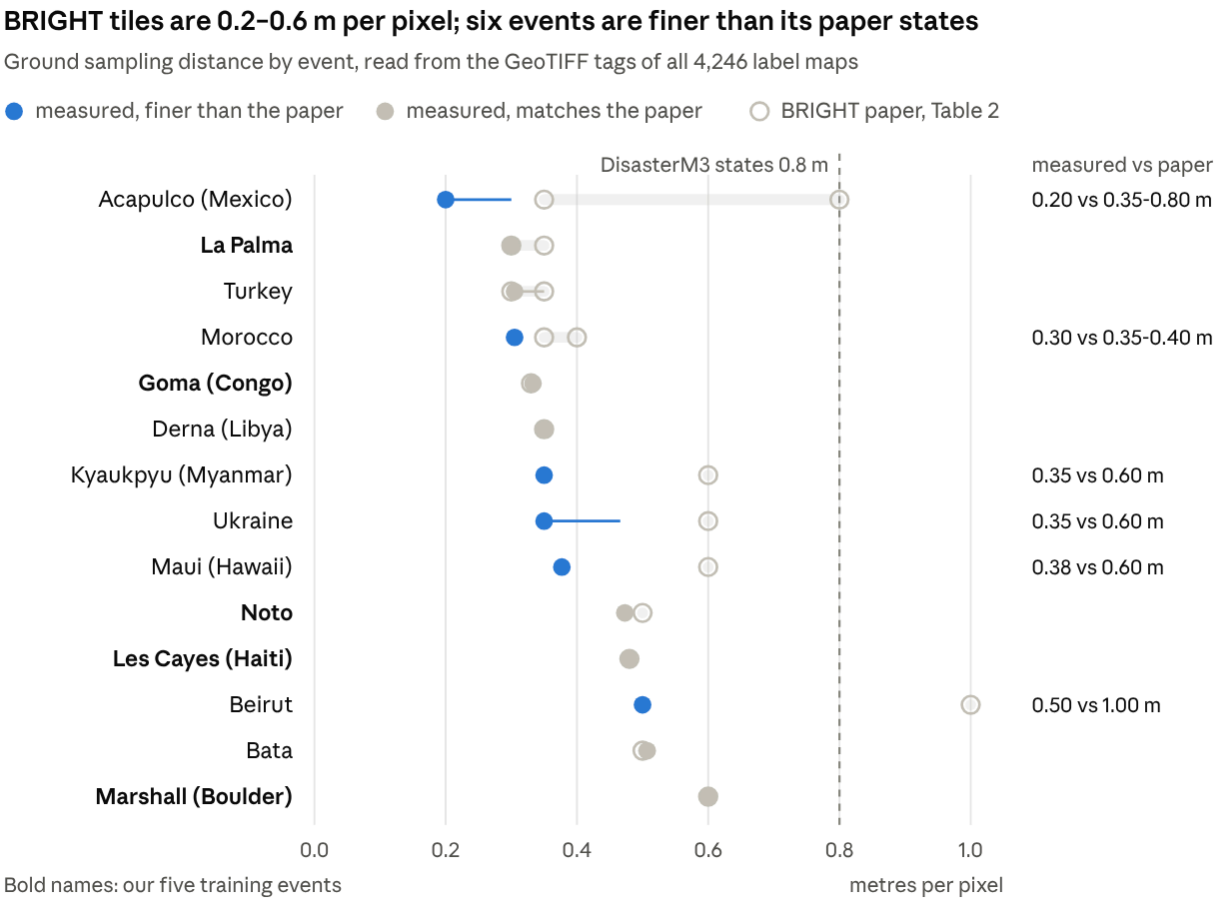
| Dataset · part | Optical pre | Optical post | SAR post (excluded) | Notes |
|---|---|---|---|---|
| DisasterM3 Instruct, files on disk | 9,746 | 7,705 | 3,537 | 20,988 files, the paper's total; the paper splits them 17,190 optical / 3,798 SAR |

| Dataset · part | Optical pre | Optical post | SAR post (excluded) | Notes |
|---|---|---|---|---|
| DisasterM3 Instruct, referenced by the JSON | 9,116 | 7,159 | 2,647 | 2,066 files are referenced by no entry |
| DisasterM3 optical pool (champion notebooks) | 6,443 | 6,443 | 0 | 5,798 train / 645 frozen val |
| DisasterM3 Bench (JSON only) | 2,679 | 2,199 | 856 | images not in the local copy; paper: 5,024 optical / 976 SAR |
| BRIGHT | 3,395 here (4,246 released) | 0 | 4,246 | Ukraine (513), Myanmar (126), Mexico (212) pre tiles not in the local copy |

- **Tags are right; file names are not.** DisasterM3's tags and the pixels agree on every referenced post. But 2,584 of the 3,537 SAR posts (73%) have no `sar` in the file name, e.g. `la_palma_volcano_00000002_post_disaster.png`. A name-based filter keeps them as optical, and the folder's older `dm3_bright_audit.py` detects SAR exactly that way.
- **Eight pre images are over 90% black** (1 `rwanda_volcano`, 4 `spain_volcano`, 3 `ukraine_conflict`). Their few data pixels are in colour, so a channel test over the whole image mistakes them for SAR; none reaches the optical pool.
- **2,066 unreferenced files** come almost entirely from BRIGHT-derived events: `spain_volcano` 478, `la_palma_volcano` 367, three spellings of Morocco 617, `libya_flood` 126.
- **36 event names**, matching the paper's 36: 19 from xBD, 14 from BRIGHT (two renamed: *Spain volcano* = La Palma, *Rwanda volcano* = Goma), and 3 others (Kalehe, Shovi, Hurricane Ian). The paper's text calls 10 events new; only those 3 come from neither source. Five events appear only in Bench: Guatemala volcano, Joplin tornado, Mexico earthquake, Hurricane Ian, Noto earthquake.

## Resolution and geometry

BRIGHT's own GeoTIFF tags put its tiles at 0.2–0.6 m per pixel, and DisasterM3's 2,828 copies of BRIGHT tiles stay at 0.30 m, not the 0.8 m its paper states.



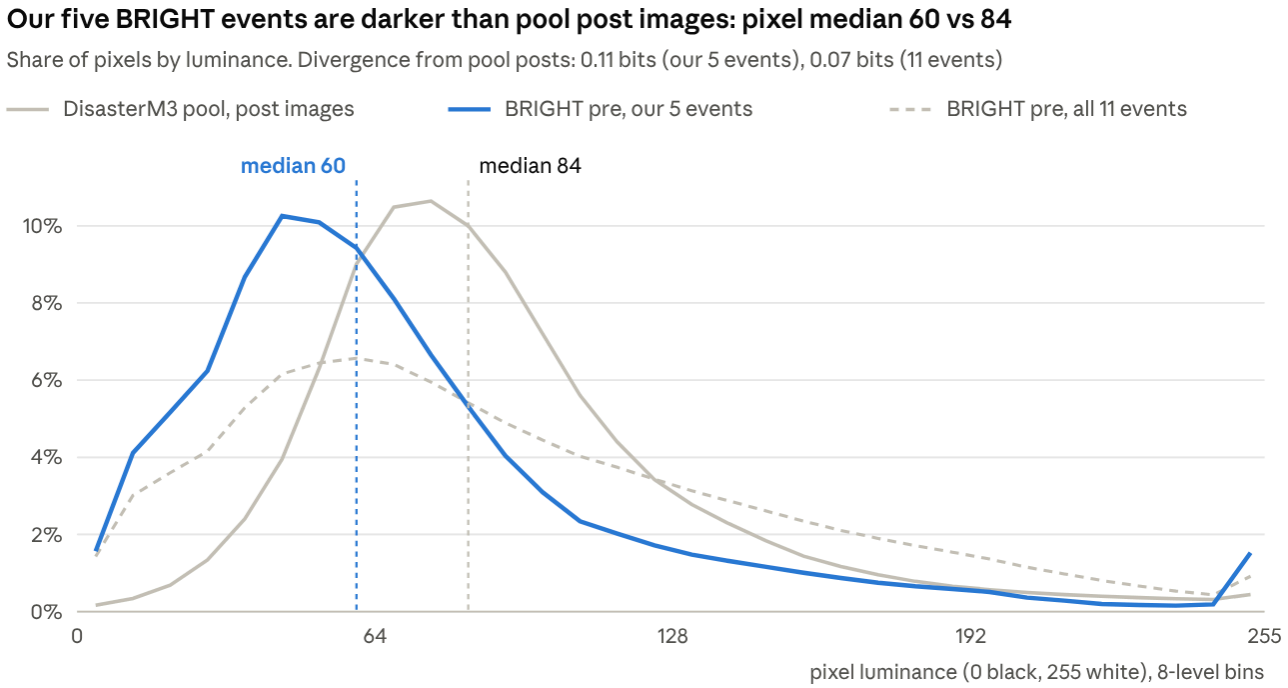

dm3_bright_optical_eda.py · GeoTIFF ModelPixelScale tags of 4,246 BRIGHT label maps (degrees converted at each tile's latitude) · BRIGHT paper Table 2 · DisasterM3 paper §3

Beirut is the widest gap: 0.50 m in the tags, 1.0 m in the paper.

- **Tile size.** Every optical image in both datasets is 1024 × 1024 px: 17,451 in DisasterM3, 3,395 in BRIGHT.
- **DisasterM3's 0.8 m.** Its §3 says all images were resampled to 0.8 m. The 2,828 DisasterM3 images that are byte-identical BRIGHT tiles have a median of 0.30 m. DisasterM3 PNGs carry no georeference, so the scale of its other images cannot be measured.
- **Sharpness.** Median high-frequency share: pool post 0.073, pool pre 0.060, BRIGHT (11 events) 0.067, our five events 0.189 (La Palma 0.223). Turkey and Morocco pool posts (0.011 and 0.009) are the blurriest images in the pool, and they are paired with BRIGHT pre tiles 3–9× sharper.
- **Object scale.** Median building component: 1,030 px in the pool, 945 px across BRIGHT, 471 px in our five events (La Palma 635 px, about 57 m²; Goma 422 px, about 47 m²).

# Radiometry and domain shift

Our five BRIGHT events are the furthest from the images the model trains and is scored on: darker, less saturated and sharper, at 0.11 bits of luminance divergence from pool post images.



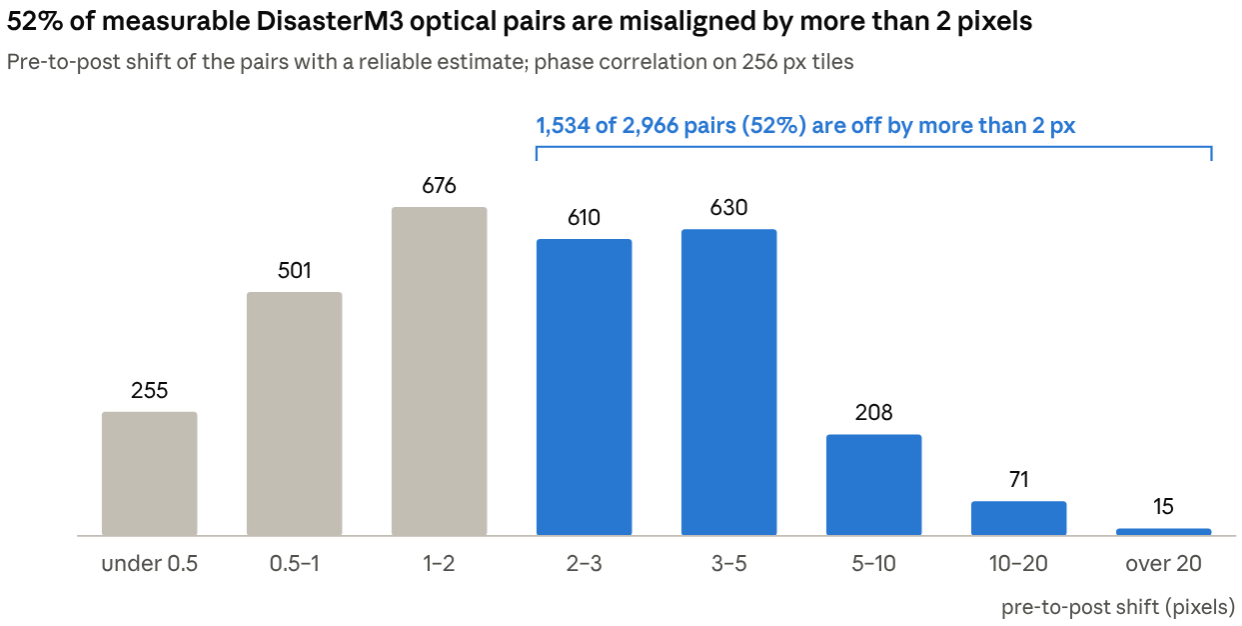

dm3_bright_optical_eda.py · luminance of non-black pixels, pooled over 6,443 pool post images, 3,395 BRIGHT pre tiles and the 1,285 tiles of our five events

Most of the gap is La Palma's dark volcanic ground (median image brightness 60). DisasterM3's own events already span 61 (Florence) to 130 (Maui) in median image brightness.

- **Pre vs post inside DisasterM3.** Pooled, pool pre and post images are almost identical (0.002 bits), but single pairs are not: median per-pair divergence 0.057 bits, and 18% of pairs exceed 0.2 bits (floods, burn scars, other sensors or seasons).
- **Colour.** Median saturation is 0.30 for pool post images and 0.21 for our five events; median image brightness 89 vs 66.
- **No-data.** 10.3% of pool images and 15.9% of BRIGHT tiles are more than 10% black. The worst BRIGHT events are Bata (28% of tiles), Goma (28%), Marshall (27%) and Derna (26%).
- **Cloud, haze, smoke.** 139 pool post images but only 15 pre images are more than 20% bright and unsaturated, so disaster-day conditions mostly hit the images the model reads.

# Pre/post registration

Half of DisasterM3's measurable optical pairs are misaligned by more than 2 px (median 2.1 px), against the paper's "strictly aligned", and the offset depends on the event.



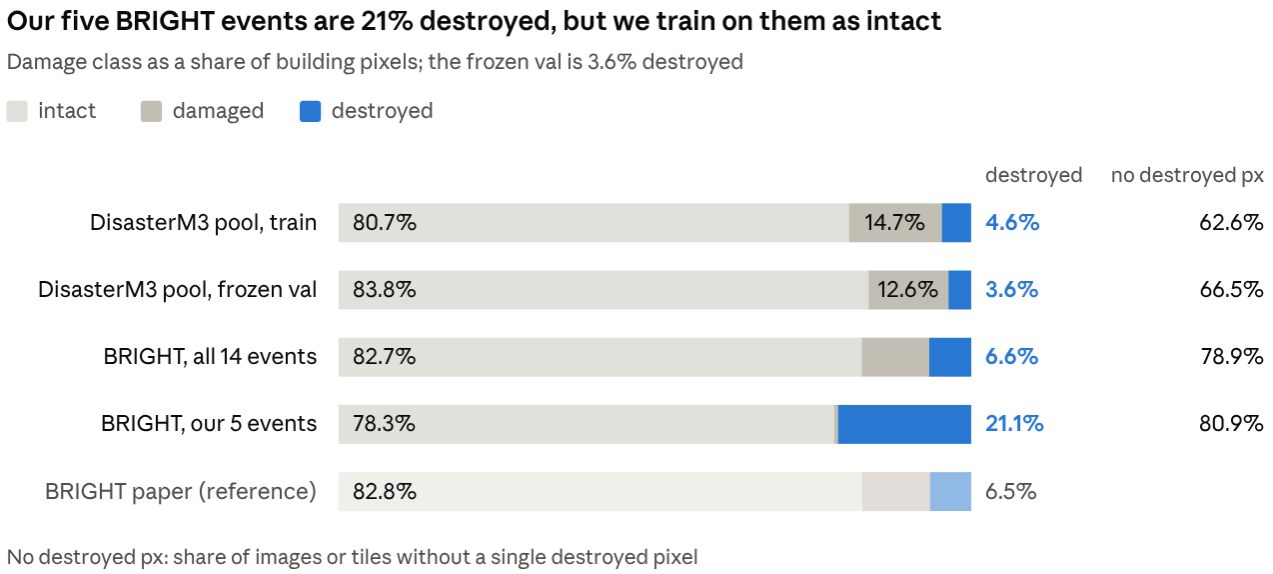

dm3_bright_optical_eda.py · 2,966 of DisasterM3's 7,159 optical pairs with a reliable shift estimate (at least 4 tiles agreeing within 1.5 px)

Portugal wildfire pairs are nearly aligned (median 0.8 px); Hurricane Harvey (7.1 px), Turkey scene 7 (10.5 px) and Morocco (18.7 px) are not.

- **Coverage.** A shift could be measured reliably for 2,966 of 7,159 optical pairs (41%). In the rest, real change, cloud or no-data stops the tiles from agreeing; 47 pairs have no tile valid in both images.
- **Worst cases.** `turkey_earthquake1` pairs 507, 516 and 522 are off by 40–56 px; `hurricane_florence` pairs 107, 150, 427 and 476 by 26–31 px.
- **Who is affected.** The champion reads only the post image, so the shift never enters it. Any pre+post model (Siamese or change detection) needs alignment or shift augmentation first.

# Labels and class balance

Destroyed buildings are scarce where the model is scored: 3.6% of building pixels in the frozen val, with 66.5% of val images holding none.




dm3_bright_optical_eda.py · 4-class maps of the 6,443 pool images (the notebooks' merge) and the 4,246 BRIGHT label maps · BRIGHT paper Fig. 5d

BRIGHT's label maps match its paper (14.1% building pixels; 82.7 / 10.7 / 6.6%). The problem is not the labels but the image: our pipeline only sees their pre-event view.

- **The merge is clean.** No pixel is claimed by two damage masks in the 6,443 pool images, and every mask is strictly 0/255, so the notebooks' merge order changes nothing.
- **Spread by event.** The destroyed share of building pixels runs from 0.07% (Turkey scene 0) to 63% (Maui) inside the pool.
- **Object counts.** The pool holds 249,623 building components (median 1,030 px). BRIGHT's maps hold 211,318 against the 384,596 buildings in its paper, because touching buildings merge (Turkey: 66,055 vs 135,033).
- **Empty tiles.** 839 BRIGHT tiles (19.8%) contain no building, 315 of them in our five events (24.5% of their tiles). No pool image is empty.
- **Roads.** Pool road pixels are 92.0% intact, 6.6% flooded and 1.4% debris-covered; the segmentation model ignores roads.

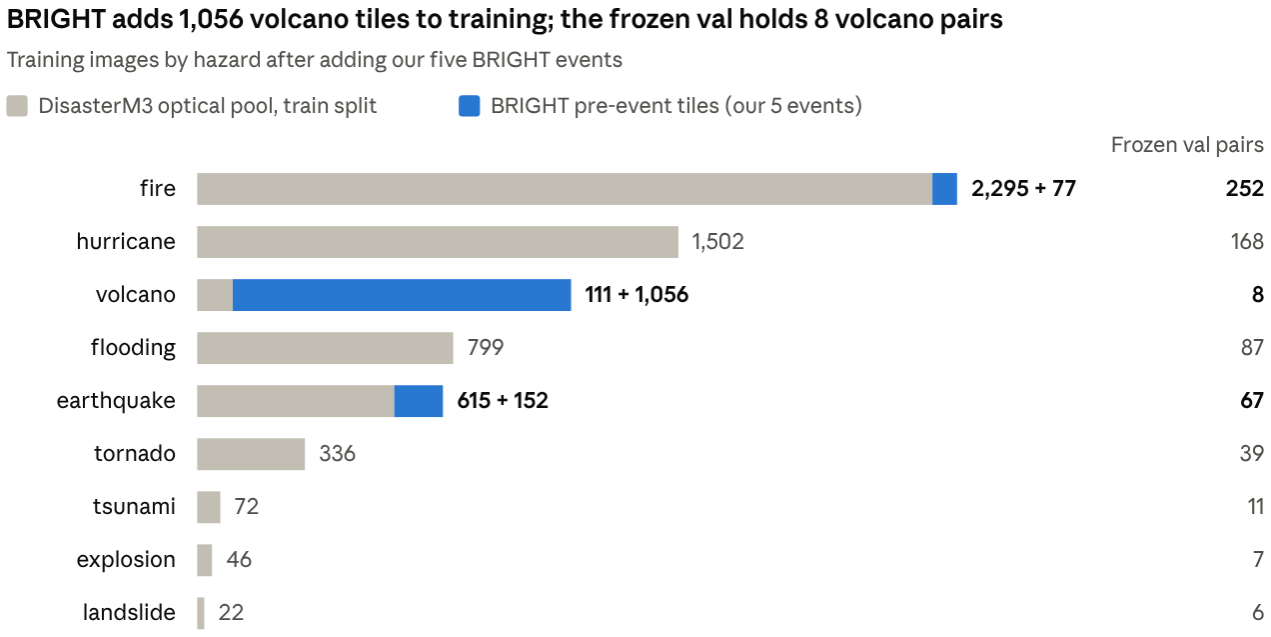
## Events and overlap

All 14 BRIGHT events are inside DisasterM3, but only six reach its optical pool, and three of those six reuse BRIGHT's pre-event tiles byte for byte.

| BRIGHT event | Name in DisasterM3 | Optical pool pairs (train / val) | SAR pairs | BRIGHT pre tiles copied byte for byte |
|---|---|---|---|---|
| Turkey earthquake | `turkey_earthquake0` – `7` | 557 / 64 | 580 | 583 of 1,114; 579 are pool pre images |
| Morocco earthquake | `morocco_earthquake`, `morroco_earthquake1` | 58 / 3 | 453 | 428 of 567; 61 are pool pre images |
| Derna flood | `libya_flood` | 43 / 4 | 51 | 52 of 124; 47 are pool pre images |
| Beirut explosion | `beriut_explosion` | 23 / 4 | 27 | 0 of 133 |
| Bata explosion | `bata_explosion` | 23 / 3 | 26 | 0 of 107 |
| Maui wildfire | `hawaii_wildfire` | 14 / 1 | 9 | 0 of 65 |
| La Palma volcano | `la_palma_volcano`, `spain_volcano` | 0 / 0 | 1,652 | 921 of 933 |
| Goma volcano | `rwanda_volcano` | 0 / 0 | 93 | 93 of 123 |
| Marshall wildfire (Boulder) | `marshall_wildfire` | 0 / 0 | 17 | 17 of 77 |
| Les Cayes earthquake | `haiti_earthquake` | 0 / 0 | 8 | 8 of 73 |
| Noto earthquake | `noto_earthquake` (Bench only) | 0 / 0 | 0 | 0 of 79 |
| Ukraine conflict | `ukraine_conflict` | 0 / 0 | 413 | not testable: no local pre tiles |
| Acapulco hurricane | `mexico_hurricane` | 0 / 0 | 112 | not testable: no local pre tiles |
| Kyaukpyu cyclone | `myanmar_hurricane` | 0 / 0 | 96 | not testable: no local pre tiles |

- **Our five events cannot leak into train or val.** They have no optical-pool pair; their copies in DisasterM3 are SAR pairs and Bench masks.

- **Turkey, Morocco and Libya would leak.** 66 frozen-val pairs use a BRIGHT tile as their pre image, so those BRIGHT tiles must stay out of training.




dm3_bright_optical_eda.py · DisasterM3 optical pool train split (5,798 pairs) and frozen val (645); BRIGHT pre tiles of La Palma, Goma, Les Cayes, Marshall, Noto (1,285)

Our five events add data almost only where the val measures least: 1,056 of their 1,285 tiles are volcano, a hazard with 8 of the 645 val pairs.

**Duplicates inside DisasterM3.** `la_palma_volcano` and `spain_volcano` store the same La Palma pairs twice: 1,456 identical images, all in SAR pairs. `socal_fire` and `woolsey_fire` share 264 identical pre images inside the optical pool; the split audit below shows what that does to the frozen val.

## Frozen-split audit

The 645-pair validation set mirrors the train split by event and class, but 40 of its pairs (6.2%) show scenes that are also in train, and they hold 8.6% of its destroyed pixels.

| Measure | Train | Frozen val |
|---|---|---|
| Pairs | 5,798 | 645 |
| From xBD events | 87.2% | 86.8% |
| Building share of all pixels | 7.21% | 7.45% |
| Damaged share of building pixels | 14.67% | 12.63% |

| Measure | Train | Frozen val |
|---|---|---|
| Destroyed share of building pixels | 4.63% | 3.59% |
| Images with no destroyed pixel | 62.6% | 66.5% |

- **How the leak happens.** xBD's `socal_fire` and `woolsey_fire` contain the same 264 pre-event tiles under two event names. The random split put 40 of them in val and their twin in train: identical pre image, post images correlated at r = 0.88 (median), building masks overlapping at IoU 0.72, and the same damage class on 97% of shared building pixels.
- **Why a name check misses it.** No val pair shares a file name with a train pair; only the pixels match.
- **What it does to the gate.** The 40 pairs hold 155,812 of the val's 1,808,111 destroyed pixels, so the 645-pair score partly rewards memorising their train twins. Report Destroyed IoU on the 605 clean pairs next to the 645 figure; the list is in `val_train_leaks.csv`.

Every val event also appears in train, as expected from a random tile split; DisasterM3 tiles carry no coordinates, so neighbouring-tile leakage cannot be measured here.

## Data-quality issues

Eleven issues were found; only the first biases the frozen validation score, and the rest change statistics, filters or preprocessing.

| Issue | Count | Effect on our pipeline |
|---|---|---|
| Same scene in train and frozen val ( `socal_fire` = `woolsey_fire` ) | 40 val pairs, 8.6% of val destroyed pixels | inflates the 645-pair Destroyed IoU |
| Pre/post misalignment | 52% of 2,966 reliably measured pairs over 2 px, 10% over 5 px, worst 55.6 px | none for the post-only champion; a pre+post model must absorb it |
| Black no-data over 10% of the tile | 10.3% of pool images, 15.9% of BRIGHT tiles | wasted crops; skews per-image normalisation |
| Upsampled post images | Turkey posts carry 3× less high-frequency energy than their pre tiles, Morocco 9× less | blurry posts beside sharp 0.30 m pre tiles |

| Issue | Count | Effect on our pipeline |
|---|---|---|
| Bright, unsaturated cover (cloud, haze, smoke) over 20% of the tile | 139 pool post images, 15 pool pre images | hides damage in the input the model sees |
| SAR posts without `sar` in the file name | 2,584 of 3,537 | name filters let SAR into optical sets |
| Same La Palma pairs under two names (`la_palma_volcano` = `spain_volcano`) | 1,456 identical images | double-counts La Palma in DisasterM3 statistics |
| Misspelt or split event names | `beriut`, `morroco`; Morocco under 3 prefixes, Turkey under 6 | event grouping by file name is wrong without a map |
| Files referenced by no JSON entry | 2,066 | invisible to the JSON-driven loaders |
| BRIGHT tiles with no building | 839 of 4,246; 315 of our 1,285 | all-background training tiles |
| Stated resolution wrong | BRIGHT Table 2 for 6 of 14 events; DisasterM3's 0.8 m for its 2,828 BRIGHT copies (0.30 m) | resampling plans based on the papers are off |

Eight pre images are over 90% black no-data; none reaches the optical pool.

## Recommendations

Fix the validation leak first; the BRIGHT addition is safe but should not be expected to move the Destroyed gate.

1. **Report the gate twice.** Keep the frozen 645 pairs for comparability with R1–R6, and add Destroyed IoU on the 605 pairs not listed in `val_train_leaks.csv`. Any new split should group pairs by pre-image hash so twins stay on one side.
2. **Keep BRIGHT Turkey, Morocco and Libya out of training.** 66 frozen-val pairs use those BRIGHT tiles as pre images. Our five events pass: none of their tiles is in the optical pool.
3. **Set expectations for the five events.** They add pre-event footprints labelled intact, 82% of them volcano, while the val holds 8 volcano pairs. Their own labels mark 21% of building pixels destroyed, but no optical post image of these events is in the local data; only Noto has one anywhere (68 in DisasterM3 Bench).

4. **Match appearance before mixing.** The five-event tiles are darker (median luminance 66 vs 89) and sharper than pool post images, and their buildings are smaller in pixels (median 471 vs 1,030 px). Normalise per image, add blur and colour jitter, and consider upsampling them about 1.5× to match object scale.
5. **Filter SAR by tag or pixels, never by file name.** Retire the name-based SAR logic in `dm3_bright_audit.py` and `dm3_vs_bright_compare.py`.
6. **Skip mostly-black crops.** 10–16% of tiles have over 10% no-data; reject random crops that are more than half black.
7. **Align pairs before any pre+post model.** The champion reads post images only, but a Siamese model would see a median 2 px shift, over 5 px in one pair in ten.
8. **Take resolution from GeoTIFF tags, not the papers.** BRIGHT's Table 2 is wrong for 6 events, and DisasterM3's BRIGHT copies are 0.30 m, not 0.8 m.

## Reproduce and definitions

Two commands regenerate every number in this audit, in about 35 minutes on an 8 GB laptop.

```
cd Essential_task_delegation/BRIGHT_DisasterM3_Integration
python dm3_bright_optical_eda.py   # audit + EDA ->
data/dm3_bright_optical_eda/
python bright_dm3_dedup.py         # per-tile BRIGHT check against the
optical pool
```

| Output file | One row per |
|---|---|
| `summary.json` | every number quoted here |
| `dm3_images.csv` | DisasterM3 Instruct image: role, event, tag, pixel modality, radiometry, md5, split |
| `bright_pre_tiles.csv` | BRIGHT pre tile: GSD, radiometry, md5 |
| `bright_targets.csv` | BRIGHT label map: class pixels, building components, GSD |
| `dm3_pool_labels.csv` | optical-pool image: class pixels, building components, road pixels |
| `dm3_optical_pair_registration.csv` | optical pair: shift, agreeing tiles, reliability |
| `dm3_pool_pairs_radiometry.csv` | pool pair: pre and post luminance, saturation, divergence |

| Output file | One row per |
|---|---|
| `dm3_events.csv` | DisasterM3 file prefix: hazard, origin, counts |
| `val_train_leaks.csv` | leaked val pair: its train twin and how close they are |

- **High-frequency share:** spectral power between half and full Nyquist in the central 512 px crop; upsampled or blurred imagery scores near 0.
- **JS divergence:** Jensen–Shannon divergence of two luminance histograms, in bits (0 identical, 1 disjoint).
- **Reliable shift:** phase correlation on 256 px tiles valid in both images; at least 4 tiles with a peak of 0.1 or more, agreeing within 1.5 px.
- **Building components:** 8-connected regions of building pixels; touching buildings merge, so dense areas are undercounted.

Corrections to the earlier PDF audit, claim by claim: Earlier PDF: corrections

# Earlier PDF: corrections

The earlier PDF (*EO Data Audit & Exploratory Data Analysis: DisasterM3 vs. BRIGHT*, 4 pages) transcribes the papers' benchmark tables correctly, but most of its dataset figures are wrong or contradict the files on disk.

| PDF section · claim | Verdict | Correct figure (measured unless marked) |
|---|---|---|
| Cover: "4,246 BRIGHT pre-event EO tiles" | Wrong for the release | 3,395 optical pre tiles; the 851 Ukraine, Myanmar and Mexico tiles are Google Earth imagery, fetched separately with `aoi_reference_kmz.zip` and BRIGHT's tutorial |
| Cover: BRIGHT "0.3–1.0 m GSD" | Wrong | 0.2–0.6 m in the GeoTIFF tags |
| Cover: "22,214 DisasterM3 EO images (0.8 m)" | Count is the paper's; GSD wrong | local Instruct: 17,451 optical images by pixel test (paper 17,190); its 2,828 BRIGHT copies are 0.30 m |
| Cover: "11,107 pure EO bi-temporal pairs" | Wrong | the PDF's own arithmetic gives 17,440 / 2 = 8,720; the JSON references 7,159 optical pairs (Instruct) + 2,199 (Bench) = 9,358 |
| Cover: "384,596 audited building footprints" | Paper figure, not audited | the paper's per-event rows sum to 358,340; the label maps hold 211,318 connected components (touching buildings merge) |
| §1: BRIGHT "5-event T1+T2 EO subset" | Not in the release | used only in BRIGHT's §4.4 experiment; all 4,246 released post tiles are SAR |
| §1: DisasterM3 "all scenes resampled to 0.80 m" | Wrong | 2,828 DisasterM3 images are byte-identical BRIGHT tiles at 0.30 m |
| §1: DisasterM3 "georeferenced to 0.8 m grid" | Wrong | DisasterM3 PNGs carry no georeference |

| PDF section · claim | Verdict | Correct figure (measured unless marked) |
|---|---|---|
| §1, §3: "xBD 15 + BRIGHT 11 + 10 new events (incl. Nepal flooding)" | Wrong | 19 xBD + all 14 BRIGHT (two renamed) + 3 new: Kalehe, Shovi, Hurricane Ian; Nepal flooding is an xBD event |
| §1: DisasterM3 "instance masks" | Imprecise | one binary 0/255 mask per class and image; instances only through connected components |
| §2: GSD column | Wrong for 7 of 14 events | Beirut 0.50 (not 1.00), Ukraine 0.35–0.47 (0.60), Myanmar 0.35 (0.60), Maui 0.38 (0.60), Morocco 0.31 (0.35–0.40), Acapulco 0.20–0.30 (0.35–0.80), Noto 0.47 (0.50) |
| §2: EO tiles for Ukraine 513, Myanmar 126, Acapulco 212 | Not released | 0 in `pre-event.zip`; DisasterM3 holds 410, 96 and 112 of them, with BRIGHT's labels |
| §2: "Reg. RMSE" column | SAR content | optical-to-SAR registration from BRIGHT Table C1; optical pre/post misregistration in DisasterM3: 52% of measurable pairs over 2 px |
| §2: "5 events (488 tiles)" with post-event EO | Wrong sum, not released | 133 + 107 + 65 + 124 + 79 = 508 tiles; no optical post image is released |
| §3: BRIGHT tiles "downsampled to 0.80 m", a 2.28–2.67× resolution loss | Wrong premise | DisasterM3 keeps them byte-identical at 0.30 m (La Palma, Goma, Les Cayes, Marshall, Turkey, Morocco, Derna) |
| §4: BRIGHT 85.9 / 14.1%, 6.1:1 (paper ~7:1) | Correct | 14.14% building pixels, 6.07:1 |
| §4: BRIGHT 82.8 / 10.7 / 6.5% | Correct | 82.7 / 10.7 / 6.6% |
| §4: DisasterM3 92.85 / 1.81 / 5.34%, "damaged collapses 5.9×" | Contradicted by the masks | optical pool: 81.0 / 14.5 / 4.5% of building pixels (5.86 / 1.05 / 0.33% of all pixels, as in our Integration Plan); damaged is higher than BRIGHT's |
| §4: roads 79.44 / 12.94 / 7.62% | Paper figure | optical-pool road pixels: 92.0 / 6.6 / 1.4% |

| PDF section · claim | Verdict | Correct figure (measured unless marked) |
|---|---|---|
| §4: 27,074 vs 26,988 "due to 86 multi-hazard tiles" | Unsupported | the paper gives no reason; 86 is only the difference |
| §4: hazard image counts (6,750 fire …) | Correct (paper Fig. 2) | they pool optical + SAR and Instruct + Bench; the optical pool is fire 2,547, hurricane 1,670, flooding 886, earthquake 682, tornado 375, volcano 119, tsunami 83, explosion 53, landslide 28, conflict 0 |
| §5A: BRIGHT Table 7 gains (F1 68.71 → 73.59 …) | Numbers right, but SAR | Table 7's post-event input is SAR, so these are optical + SAR results |
| §6: BRIGHT Table 8 and DisasterM3 Tables 2, 3, 9 | Correct | every value checked against the papers; one label is wrong: BRIGHT rows put Intact IoU under "Counting" |
| §7: +3.80 mIoU, +8.32 Destroyed IoU, +5.7 mIoU, +7.9 Building IoU, 5.4× fewer parameters | Correct | checked against BRIGHT Table 8 and DisasterM3 Table 3 |
| Pages 2–4 | Text cut off | Sections 3, 5A and 7 end mid-sentence |

## Missing from the PDF

- The frozen-val leak: 40 pairs whose `socal_fire` / `woolsey_fire` twins sit in train, holding 8.6% of val destroyed pixels.
- All 14 BRIGHT events live inside DisasterM3, and BRIGHT's Turkey, Morocco and Libya tiles are the pre images of 66 val pairs.
- 73% of DisasterM3's SAR posts have no `sar` in the file name.
- Pre/post misregistration and the appearance gap of the BRIGHT tiles.

## Suggested fixes

1. Replace Sections 1–5 with the measured figures in the main tab, keeping "paper says" and "measured" in separate columns.
2. Keep Sections 6–7, fixing the column label.

3. Drop the SAR-based content (Table 7, Table C1), or label it optical + SAR.
4. Put the table or figure number beside every paper figure.
5. Re-export so no section is cut off.